Multilingual Multimodal Hate Speech Detection Using LLM-Guided Cross-Lingual Reasoning

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Abstract

Hate speech detection on social media has become increasingly challenging due to the widespread use of multilingual text, images, and memes to convey harmful content. Traditional systems primarily focus on text in a single language and often fail to capture the contextual meaning when multiple modalities are involved.

In this work, we propose a multilingual multimodal hate speech detection framework that integrates textual and visual information to improve classification performance. The system leverages a multilingual language model for text understanding and a convolutional neural network for extracting visual features. Additionally, an explanation module is incorporated to generate human-readable reasoning for model predictions, improving interpretability.

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1. Introduction

With the rapid growth of social media platforms, the volume of harmful and offensive content has increased significantly. Detecting such content automatically is a critical challenge, especially when users communicate using multiple languages and combine text with images.

During initial analysis, it was observed that certain content appears non-offensive when viewed in isolation but conveys harmful intent when both text and visual context are considered together. This is particularly common in memes and multilingual or code-mixed content.

Another major limitation of existing systems is the lack of interpretability. Most models act as black boxes, providing predictions without explaining the reasoning behind them. This reduces trust and limits their practical applicability.

To address these challenges, we propose a system that combines multilingual text understanding, multimodal feature integration, and explanation generation.

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2. Proposed Methodology

2.1 System Overview

The proposed system takes both textual input (in multiple languages) and corresponding images as input and classifies the content as hateful or non-hateful. In addition to classification, the system generates a brief explanation for each prediction.

2.2 Implementation Plan

To ensure a structured approach, the system will be developed in the following stages:

- Phase 1: Dataset collection from multilingual and multimodal sources (including meme-based datasets)
- Phase 2: Development of a baseline text-only model using multilingual BERT (mBERT)
- Phase 3: Extraction of visual features using ResNet-50
- Phase 4: Multimodal fusion of text and image representations
- Phase 5: Model training and evaluation using standard metrics (F1-score, precision, recall)
- Phase 6: Integration of an LLM-based explanation module

2.3 Model Components

- **Text Encoder:** Multilingual BERT (mBERT) is used to capture contextual representations across different languages
- **Image Encoder:** ResNet-50 is used to extract visual features from images
- **Fusion Layer:** Combines textual and visual features to capture cross-modal relationships
- **Classifier:** Predicts whether the content is hateful or not
- **Explanation Module:** Generates simple natural language explanations for model predictions

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2.4 Training Details (Planned)

- Optimizer: AdamW

- Learning Rate: 2e-5
- Maximum sequence length: 128
- Batch size: 32

During training, challenges such as class imbalance and multilingual variability will be addressed using appropriate techniques such as weighted sampling.

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3. Datasets

The system will be trained and evaluated on multilingual datasets covering languages such as English, Hindi, etc. For multimodal analysis, meme-based datasets containing both text and image inputs will be used.

These datasets enable evaluation across different linguistic and contextual scenarios.

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4. Expected Results and Discussion

The proposed system is expected to outperform unimodal approaches (text-only or image-only models) by effectively capturing cross-modal context.

Preliminary observations suggest that combining textual and visual features significantly improves detection accuracy. The inclusion of explanation generation further enhances usability by providing interpretable insights into model decisions.

However, challenges such as sarcasm detection, code-mixed language understanding, and ambiguous visual context are expected to impact performance and will be explored further.

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5. Limitations

- Dependence on pretrained models may introduce bias
- Explanation generation may not always perfectly reflect internal model reasoning
- Performance may vary across languages due to dataset imbalance

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6. Conclusion

This work proposes a multilingual multimodal framework for hate speech detection that integrates text and image understanding with explanation generation. By combining these

components, the system aims to improve both classification performance and interpretability, making it more suitable for real-world deployment.

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7. Future Work

Future improvements include:

- Extending support to additional languages
- Improving explanation accuracy and faithfulness
- Handling sarcasm and indirect hate speech more effectively
- Exploring real-time deployment scenarios